

Hand-Crafted Features, Fine-Tuned CNNs and Their Ensembles for Aerial Scene Classification under Class Imbalance

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Abstract—Aerial scene classifiers have to tell apart land-use classes that look alike, and they are usually trained on collections where some classes are rare. We revisit a standard comparison on the 15-class SkyView dataset: hand-crafted pipelines (local binary patterns, colour histograms and SIFT bag-of-words with k -nearest-neighbour, SVM and stacking classifiers) against fine-tuned ResNet-18 and EfficientNet-B3, plus voting ensembles of both. Every model is tuned on the same validation set and scored on the same balanced test set, and we train on three sets that differ only in their class distribution: balanced, long-tailed (16:1) and rebalanced by offline augmentation. With balanced training the CNNs reach 96.2% and 96.5% macro-F1, against 84.4% for the best hand-crafted pipeline. The long tail costs the CNNs 2.4–4.0 points and the hand-crafted pipelines 9–13; stacking recognises only 2.5% of the rarest class. Rebalancing by augmentation helps the hand-crafted pipelines but lowers both CNNs by 1.1–1.7 points, while weighting the loss by inverse class frequency raises them by 0.9–1.2. Averaging the two CNNs beats the better network in every setting (98.0% macro-F1 on balanced data, McNemar $p < 10^{-4}$); adding hand-crafted members does not help. Finally, we show that breaking voting ties by member order turns any two-member majority vote into a copy of its first member, which is why these ensembles previously appeared to collapse.

Index Terms—aerial scene classification, remote sensing, transfer learning, class imbalance, ensemble learning, Grad-CAM

I. INTRODUCTION

Aerial scene classification assigns a land-use or land-cover label, such as *Port*, *Railway* or *River*, to an overhead image, and it underpins many mapping and monitoring tasks. Two things make it hard. Classes that share colour and texture must be told apart: on the SkyView data used here, a hand-crafted pipeline most often mistakes *Highway* for *Railway* and *Lake* for *River*. And real collections are rarely balanced, so some classes come with far fewer training images than others [1].

The usual comparison pits two families of methods against each other. Hand-crafted pipelines summarise an image with colour, texture or keypoint statistics and train a shallow classifier on top [2]–[5], while ImageNet-pretrained CNNs are fine-tuned end to end [6], [7]. Ensembles are often suggested as a way to get the best of both [8], [9], and augmentation or loss re-weighting is the standard response to imbalance [10], [11]. These comparisons are easy to get wrong, though. If each method is tuned on its own split, or hyperparameters are

chosen on the test images, a results table no longer separates the effect of the method from the effect of the protocol.

We ran into exactly this problem. An earlier version of this study, a course project, tuned on the test set, mixed numbers from different splits, and broke voting ties by member order. That last choice is subtle: with two voters every disagreement is a tie, so a vote that favours the first-listed member simply returns that member’s predictions. The ensembles looked as if they collapsed to their weakest member when in fact they were never combining anything.

This paper repeats the comparison under a protocol designed to avoid these traps. We hold out one balanced test set and one balanced validation set, and derive three training sets from the remaining images that differ only in class distribution: balanced, long-tailed with an imbalance ratio of 16, and rebalanced by augmenting the minority classes (Section III). Every hyperparameter and stopping epoch is chosen on the validation set. On each training set we train four hand-crafted pipelines and two CNNs, ResNet-18 and EfficientNet-B3, the latter with a linear-probe-then-fine-tune schedule (Section IV), and we combine them by soft voting and by hard voting with a probability-based tie-break. All comparisons are made on the same 2,400 test images, with bootstrap confidence intervals and paired McNemar tests.

Our contributions are:

- a controlled protocol and re-implementation for comparing hand-crafted features, fine-tuned CNNs, ensembles and rebalancing strategies on SkyView (Sections III and IV);
- measurements of how far apart the two families are on balanced data and how much each loses under a 16:1 long tail, broken down by head, middle and tail classes (Sections V-A and V-B);
- evidence that averaging the two CNNs significantly improves on either network in all three settings while adding hand-crafted members does not, together with a diagnosis of the tie-breaking pitfall (Sections IV-C and V-C);
- a comparison of minority-class augmentation with inverse-frequency loss weighting, which shows that augmentation helps the hand-crafted pipelines but not the CNNs (Section V-B).

II. RELATED WORK

Hand-crafted descriptors. Before learned features, aerial scene classification relied on global and local image statistics. Colour histograms [2] summarise the chromatic distribution of an image and are invariant to rotation and translation, but scenes that share a palette, such as beaches and deserts, become hard to separate. Local binary patterns (LBP) [3] encode micro-texture in a gray-scale and rotation invariant way, and SIFT [4] keypoints quantised into a bag of visual words (BoW) capture local structure; Yang and Newsam [5] applied BoW with spatial extensions to land-use classification. Such descriptors are typically paired with support vector machines (SVMs) [12] or nearest-neighbour rules [13]. Larger benchmarks such as NWPU-RESISC45 [14] and AID [15] were introduced because accuracy on earlier, smaller datasets had saturated; both papers review hand-crafted and deep methods side by side.

Transfer learning with CNNs. Convolutional networks pre-trained on ImageNet [16] are the default starting point for remote sensing scenes. Nogueira et al. [6] compared full training, fine-tuning and fixed feature extraction on three remote sensing datasets and found fine-tuning to be the strongest strategy, and Pires de Lima and Marfurt [7] reviewed transfer learning for scene classification across datasets and architectures. Kornblith et al. [17] reported a strong correlation between ImageNet accuracy and transfer accuracy over 12 datasets. We use residual networks [18] and compound-scaled EfficientNets [19] as two backbones of different capacity under one training recipe.

Ensembles and diversity. Combining classifiers reduces error when the members are accurate and err on different inputs [8], [9], [20]. Stacking [21] learns the combination with a meta-classifier, whereas voting uses a fixed rule; averaging the predictions of independently trained networks is a common deep-learning practice [22], [23]. Kuncheva and Whitaker [24] studied how measures of diversity relate to ensemble accuracy; we use one such pairwise measure, Yule’s Q -statistic, to explain when our ensembles help.

Class imbalance. Imbalanced training data degrade classifiers [1], and Buda et al. [10] found the effect to be detrimental for CNNs on MNIST, CIFAR-10 and ImageNet. Remedies include re-sampling, cost-sensitive losses such as the class-balanced [25] and focal [26] losses, and data augmentation [27]; Zhang et al. [11] survey them for deep long-tailed learning. We compare offline augmentation of minority classes with inverse-frequency loss weighting on a controlled long-tailed split.

Explainability. Grad-CAM [28] localises the image regions that drive a prediction by weighting the last convolutional feature maps with class-specific gradients. We use it to inspect the most frequent confusion of each network.

III. DATA AND EVALUATION PROTOCOL

Figure 1 summarises the study. All models, in every setting, share one split, one model-selection rule and one test set, so

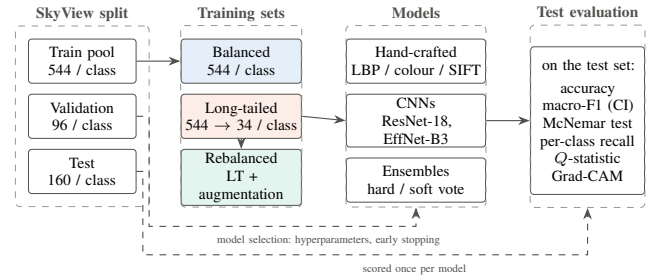


Fig. 1. Experimental workflow. A fixed balanced test set and a fixed balanced validation set are held out once; three training sets are derived from the remaining images. Every hyperparameter and early-stopping decision uses the validation set only (dashed), and the test set is scored once per model.

that differences between rows of a table can be attributed to the model or to the training distribution and not to the evaluation.

A. Dataset and splits

We use the SkyView aerial landscape dataset [29]: 12,000 RGB images of 256×256 pixels, 800 per class, in 15 classes (Agriculture, Airport, Beach, City, Desert, Forest, Grassland, Highway, Lake, Mountain, Parking, Port, Railway, Residential, River). We hold out a stratified test set of 160 images per class (2,400 images). From the remaining 640 images per class we draw a validation set of 96 images per class (1,440 images, seed 0), leaving a training pool of 544 images per class. We verified that no file appears in more than one split and that the test set contains no byte-identical copy of a training image.

B. Training distributions

We derive three training sets from the pool (Table I).

Balanced: all 8,160 pool images.

Long-tailed: class i in alphabetical order keeps a fraction r_i of its 544 pool images, with $r = (800, 700, 650, \dots, 100, 50)/800$, i.e. from 544 images for Agriculture down to 34 for River. The imbalance ratio is 16 and the set has 4,114 images.

Rebalanced: the long-tailed set plus offline augmented copies of each minority class, generated only from that class’s own long-tailed images, until every class has 544 images (4,046 synthetic images). Each copy applies a random rotation in $[-30^\circ, 30^\circ]$, Gaussian blur with radius in $[0.1, 2]$, and brightness, contrast and colour-saturation factors in $[0.8, 1.2]$.

Validation and test sets stay balanced and identical in all settings. A balanced validation set makes model selection target macro-averaged performance and keeps selection noise identical across settings, so the long-tailed and rebalanced columns differ from the balanced column only in the training data.

C. Metrics and statistical testing

We report accuracy and macro-averaged F1 on the test set; on a balanced test set accuracy equals macro-recall. For imbalance we also report per-class recall and its mean over

TABLE I
TRAINING SETS. VALIDATION (96/CLASS) AND TEST (160/CLASS) SETS
ARE SHARED AND BALANCED.

Setting	Images	Per class	Synthetic
Balanced	8,160	544	0
Long-tailed	4,114	544-34	0
Rebalanced	8,160	544	4,046

the five head, five middle and five tail classes of the long-tailed ordering. Uncertainty is quantified with 95% percentile bootstrap intervals over test images (2,000 resamples). Paired comparisons between two classifiers on the same test images use the exact McNemar test [30], and we call a difference significant at $p < 0.05$. All results come from a single training run per configuration (seed 0); the bootstrap intervals capture test-sampling variability but not training variability.

IV. METHODS

All hyperparameters are chosen with Optuna [31] using the tree-structured Parzen estimator [32], maximising validation macro-F1; the selected configuration is then refitted on the training set and applied once to the test set. Classical models use scikit-learn [33] and CNNs use PyTorch [34].

A. Hand-crafted pipelines

We extract three descriptors from each 256×256 image. *LBP*: uniform rotation-invariant patterns with $P=24$ neighbours at radius $R=3$ on the gray-scale image, giving a normalised 26-bin histogram [3]. *Colour*: a 32-bin histogram per RGB channel, each normalised to unit sum (96 dimensions) [2]. *SIFT-BoW*: up to 300 SIFT descriptors per image [4], quantised against a 100-word vocabulary learned with mini-batch k -means on 2×10^5 descriptors sampled from the training set of the current setting, giving a normalised 100-bin histogram.

On these features we train four classifiers: (i) k NN on LBP, tuning $k \in [1, 30]$, uniform or distance weighting, and the Euclidean or Manhattan metric (30 trials); (ii) an RBF-kernel SVM on standardised colour histograms, tuning $C \in [10^{-2}, 10^3]$ and $\gamma \in [10^{-4}, 1]$ (30 trials); (iii) the same SVM on SIFT-BoW; (iv) a stacking ensemble [21] on the standardised concatenation of all three descriptors (222 dimensions), whose base learners are an RBF SVM, a distance-weighted k NN and a decision tree, combined by a logistic-regression meta-learner trained on 5-fold out-of-fold base-learner outputs; we tune the SVM C , k and the tree depth (15 trials). SVM probabilities for voting come from Platt scaling.

B. CNN transfer learning

We fine-tune ImageNet-pretrained ResNet-18 [18] and EfficientNet-B3 [19] at 224×224 input with ImageNet normalisation, using a two-stage linear-probe-then-fine-tune schedule. In stage 1 the backbone is frozen and a linear head (preceded by dropout 0.3 for EfficientNet-B3) is trained on cached pooled features for 15 epochs; Optuna selects its learning rate

in $[10^{-5}, 10^{-2}]$ and weight decay w in $[10^{-6}, 5 \times 10^{-3}]$ over 20 trials. In stage 2 we unfreeze all layers of ResNet-18 and, for EfficientNet-B3, the last MBConv stage and the final 1×1 convolution, keeping the remaining BatchNorm statistics fixed. Training uses Adam [35] with weight decay w , learning rate 10^{-4} for the backbone and $\min(\eta^*, 10^{-3})$ for the head, where η^* is the stage-1 learning rate, cosine annealing [36] over at most 15 epochs, batches of 64, random horizontal flips and rotations in $[-15^\circ, 15^\circ]$. We stop after 4 epochs without a validation macro-F1 gain and restore the best epoch. In the long-tailed setting we additionally train both networks with inverse-frequency weighted cross-entropy, $w_c = N/(Kn_c)$ for K classes and n_c training images in class c , as a cost-sensitive alternative to augmentation [1], [25].

C. Ensembles

We combine base models with two fixed rules. *Soft voting* averages the members' class-probability vectors and predicts the arg-max. *Hard voting* predicts the class with the most member votes. With an even number of members ties are frequent; with two members every disagreement is a tie. If ties are broken by member order, a two-member hard vote reproduces its first member's prediction exactly, so we break ties by the summed probability of the tied classes. For each ensemble we also report the oracle accuracy, the fraction of test images that at least one member classifies correctly, as an upper bound on what any combiner could achieve. To relate ensemble gains to diversity we compute Yule's Q -statistic for each pair of classifiers [24],

$$Q_{ab} = \frac{N^{11}N^{00} - N^{10}N^{01}}{N^{11}N^{00} + N^{10}N^{01}}, \quad (1)$$

where N^{11} counts test images both classifiers get right, N^{00} those both get wrong, and N^{10} , N^{01} those only one gets right. Q ranges from -1 to 1 ; values near 1 indicate that the two classifiers fail on the same images.

D. Grad-CAM

For both networks we compute Grad-CAM [28] maps from the last convolutional block (`layer4` for ResNet-18, the final 1×1 convolution for EfficientNet-B3) with respect to the predicted class, upsampled to the input size. We visualise test images from the most frequent off-diagonal confusion of ResNet-18.

V. RESULTS

Table II reports every base model under the three training distributions and Fig. 2 shows the same macro-F1 values with their confidence intervals. Table IV reports the ensembles and Table III breaks recall down by head, middle and tail classes.

A. Balanced training

The two families are far apart. The single hand-crafted pipelines reach 53.3% (LBP + k NN), 62.9% (SIFT-BoW + SVM) and 69.8% (colour histogram + SVM) macro-F1, and stacking their three descriptors raises this to 84.4%. Both CNNs exceed 96%: ResNet-18 reaches 96.2% and

TABLE II
TEST ACCURACY AND MACRO-F1 (%) OF THE BASE MODELS; BRACKETS GIVE 95% BOOTSTRAP INTERVALS FOR MACRO-F1. ALL ROWS ARE SCORED ON THE SAME 2,400 BALANCED TEST IMAGES. THE WEIGHTED-CE ROWS TRAIN ON THE LONG-TAILED SET WITH INVERSE-FREQUENCY CLASS WEIGHTS.

Model	Balanced		Long-tailed		Rebalanced	
	Acc.	Macro-F1	Acc.	Macro-F1	Acc.	Macro-F1
LBP + k NN	53.8	53.3 [51.3, 55.2]	46.4	44.2 [42.3, 45.9]	47.5	45.6 [43.7, 47.4]
Colour hist. + SVM	70.1	69.8 [68.0, 71.5]	59.2	56.9 [55.1, 58.6]	63.8	62.0 [60.2, 63.6]
SIFT-BoW + SVM	63.3	62.9 [61.1, 64.7]	55.5	53.5 [51.7, 55.2]	56.5	54.5 [52.6, 56.4]
Stacking (SVM, k NN, DT)	84.4	84.4 [82.9, 85.8]	75.6	73.1 [71.5, 74.6]	76.8	75.4 [73.7, 76.9]
ResNet-18	96.2	96.2 [95.4, 97.0]	94.0	93.8 [92.8, 94.8]	92.5	92.1 [91.0, 93.2]
EfficientNet-B3	96.5	96.5 [95.8, 97.2]	92.6	92.5 [91.5, 93.6]	91.5	91.5 [90.3, 92.6]
ResNet-18 (weighted CE)	–	–	95.1	95.0 [94.1, 95.8]	–	–
EfficientNet-B3 (weighted CE)	–	–	93.5	93.4 [92.5, 94.4]	–	–

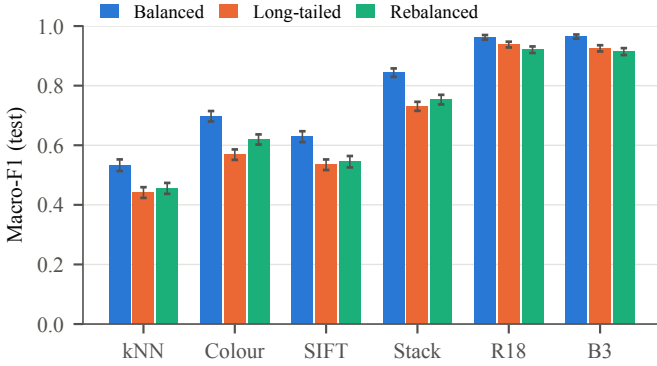


Fig. 2. CNNs lose 2.4–4.0 pp of macro-F1 under the long tail, hand-crafted pipelines 9–13 pp; rebalancing by augmentation helps only the latter. Bars show test macro-F1 of each base model under the three training sets, whiskers 95% bootstrap intervals.

EfficientNet-B3 96.5%, a difference that is not significant (McNemar $p = 0.63$; 57 test images are classified correctly only by EfficientNet-B3 and 51 only by ResNet-18). The confusion matrices in Fig. 3 locate the remaining hand-crafted errors among classes with similar texture, most often *Highway* predicted as *Railway* (11% of *Highway* images) and *Lake* as *River* (9%), whereas EfficientNet-B3 makes no off-diagonal error above 5% of a class.

B. Long-tailed training and rebalancing

Training on the long-tailed set lowers every model significantly ($p < 10^{-4}$ for each base model, McNemar against its balanced counterpart), but by very different amounts. Macro-F1 falls by 9.1 to 12.9 pp for the hand-crafted pipelines and by 11.3 pp for stacking, against 2.4 pp for ResNet-18 and 4.0 pp for EfficientNet-B3 (Table II). Under the long tail ResNet-18 is the stronger network (93.8% vs. 92.5%, $p = 0.017$).

Table III shows where the loss occurs. Head-class recall is unchanged or slightly higher than under balanced training, consistent with the shifted class prior favouring the head, while tail-class recall falls from 82.8% to 53.5% for stacking and by about 8 and 11 pp for ResNet-18 and EfficientNet-B3. The rarest class is the extreme case (Fig. 4): with 34 training

images, stacking recognises 2.5% of the *River* test images, ResNet-18 70.0% and EfficientNet-B3 76.9%.

Offline augmentation of the minority classes affects the two families in opposite directions. For the hand-crafted pipelines it recovers part of the loss: colour-histogram SVM gains 5.0 pp ($p < 10^{-4}$), LBP + k NN 1.4 pp ($p = 0.049$), and SIFT-BoW and stacking gain 1.0 and 2.3 pp without reaching significance ($p = 0.33$ and 0.14); stacking’s tail recall rises from 53.5% to 62.2%. For the CNNs it is harmful: macro-F1 falls by 1.7 pp for ResNet-18 ($p = 0.003$) and 1.1 pp for EfficientNet-B3 ($p = 0.029$), and their tail recall falls further, to 82.5% and 84.2%. *River* recall of ResNet-18 drops from 70.0% to 53.7%. Re-weighting the loss instead of augmenting the data improves both networks over plain long-tailed training, to 95.0% macro-F1 for ResNet-18 (+1.2 pp, $p = 0.003$) and 93.4% for EfficientNet-B3 (+0.9 pp, $p = 0.006$), mainly through the tail classes (91.5% and 89.8% tail recall).

C. Ensembles

Averaging the two CNNs is the only combination that improves on its best member in every setting (Table IV). The ResNet-18 + EfficientNet-B3 soft vote reaches 98.0%, 95.3% and 94.1% macro-F1 on the balanced, long-tailed and rebalanced sets, gains of 1.5, 1.5 and 2.0 pp over the better network ($p < 10^{-4}$ in each case). On the balanced test set it corrects 48 images that EfficientNet-B3 gets wrong while introducing 11 new errors.

Adding hand-crafted members does not help. Pairing a CNN with k NN or stacking leaves the CNN’s score unchanged or lowers it; under the long tail the k NN pairs lose 1.2 and 1.5 pp ($p < 10^{-4}$). The four-member ensemble of k NN, stacking and both CNNs gains 0.9 pp on the balanced and rebalanced sets but remains below the two-CNN vote, and the six-member ensemble is never significantly better than its best member and is 1.9 pp worse than ResNet-18 under the long tail ($p < 10^{-3}$). Hard voting is close to soft voting for two members, where our tie-breaking reduces it to a probability comparison, but falls well behind once four weak voters can outvote the two CNNs (88.6% vs. 92.0% for all six members under the long tail). For reference, applying the first-member tie-break instead

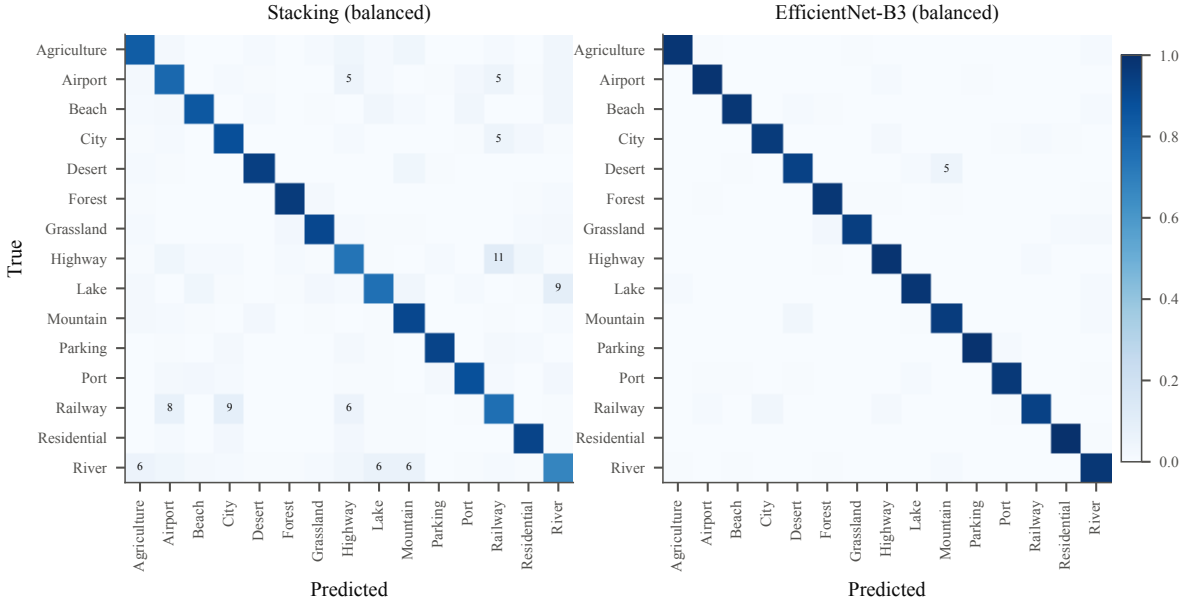


Fig. 3. Hand-crafted errors concentrate on texture-similar classes. Row-normalised test confusion matrices after balanced training for stacking (left) and EfficientNet-B3 (right); off-diagonal cells of at least 5% are annotated.

TABLE III
MEAN TEST RECALL (%) OVER THE FIVE HEAD (544–374 LONG-TAILED TRAINING IMAGES), MIDDLE (340–204) AND TAIL (170–34) CLASSES.

Model	Balanced			Long-tailed			Rebalanced		
	Head	Mid.	Tail	Head	Mid.	Tail	Head	Mid.	Tail
Stacking (SVM, k NN, DT)	85.4	85.1	82.8	90.8	82.5	53.5	87.0	81.0	62.2
ResNet-18	96.1	97.0	95.6	97.0	97.3	87.6	97.8	97.1	82.5
EfficientNet-B3	96.2	96.4	96.9	96.6	94.9	86.2	96.4	94.0	84.2
ResNet-18 (weighted CE)	–	–	–	96.2	97.5	91.5	–	–	–
EfficientNet-B3 (weighted CE)	–	–	–	95.5	95.1	89.8	–	–	–

reproduces the first member’s scores exactly for every two-member ensemble in Table IV.

The pairwise Q -statistics (Eq. (1)) show that the helpful pair is not the most diverse one. ResNet-18 and EfficientNet-B3 have the highest Q in every setting (0.93, 0.90 and 0.86), i.e. they tend to fail on the same images, while pairs of a CNN and a hand-crafted model have Q between 0.55 and 0.86. The oracle accuracy of the six-member ensemble (99.2% on the balanced set) shows that the weak members do contribute correct answers that the CNNs miss, but a fixed voting rule cannot pick them out from the many images on which the weak members are wrong and the CNNs are right.

D. Grad-CAM case study

The most frequent confusion of ResNet-18 after long-tailed training is *Railway* predicted as *Airport* (9 test images; EfficientNet-B3 makes the same error on 7). Figure 5 shows a correctly classified *Railway* image and three of these errors. On the correct image both networks attend to the fan of converging tracks. In the first error ResNet-18 concentrates on a cluster of large roofs between the tracks and predicts *Airport*, while EfficientNet-B3 follows the track lines and is correct; in the third, both networks attend to a long straight

corridor and predict *Airport*. These maps are consistent with the networks associating large paved structures and long linear strips with airports, a cue that railway scenes share; we did not test this interpretation further.

VI. DISCUSSION

Why the CNN pair helps and mixed ensembles do not. A vote can only fix an error if enough members get that image right. The two CNNs are both accurate, and although their Q is high they still disagree on a fair number of images (57 and 51 one-sided errors on the balanced set), so averaging settles most disagreements in favour of the right class. A hand-crafted member is diverse in the wrong way. It is wrong on many images the CNNs already handle, so in a pair it can only pull the stronger model down, and in a larger vote it can outvote it. This fits the familiar view that ensemble members need to be both accurate and diverse [8], [24]; here the accuracy gap between the families outweighs whatever the weak members add. A combiner learned on validation predictions might exploit the high oracle accuracy, but we have not tried one.



Fig. 4. Recall collapses on the rarest classes for stacking, and augmentation-based rebalancing does not restore the CNNs’ tail recall. Per-class test recall after long-tailed (left) and rebalanced (right) training; classes are ordered by their number of real training images.

TABLE IV

MACRO-F1 (%) OF HARD AND SOFT VOTING ENSEMBLES AND THE CHANGE OF THE SOFT VOTE RELATIVE TO ITS BEST MEMBER (Δ , PERCENTAGE POINTS); * MARKS A SIGNIFICANT DIFFERENCE (MCNEMAR, $p < 0.05$). R18: RESNET-18, B3: EFFICIENTNET-B3, STACK: STACKING ENSEMBLE.

Members	Balanced			Long-tailed			Rebalanced		
	Hard	Soft	Δ	Hard	Soft	Δ	Hard	Soft	Δ
k NN + Stack	83.2	83.2	-1.2*	70.3	70.2	-2.9*	74.3	74.0	-1.4
k NN + R18	95.9	96.0	-0.3	92.7	92.7	-1.2*	91.9	91.9	-0.2
k NN + B3	96.1	96.2	-0.3	90.9	91.0	-1.5*	91.2	91.1	-0.4
Stack + R18	96.6	96.7	+0.4	93.5	93.5	-0.3	91.7	91.7	-0.4
Stack + B3	96.8	96.9	+0.4	92.4	92.3	-0.2	91.8	91.8	+0.3
R18 + B3	98.0	98.0	+1.5*	95.3	95.3	+1.5*	94.1	94.1	+2.0*
k NN + Stack + R18 + B3	97.0	97.4	+0.9*	93.5	94.0	+0.1	92.7	93.0	+0.9*
All six	95.5	96.9	+0.4	88.6	92.0	-1.9*	89.1	91.3	-0.8

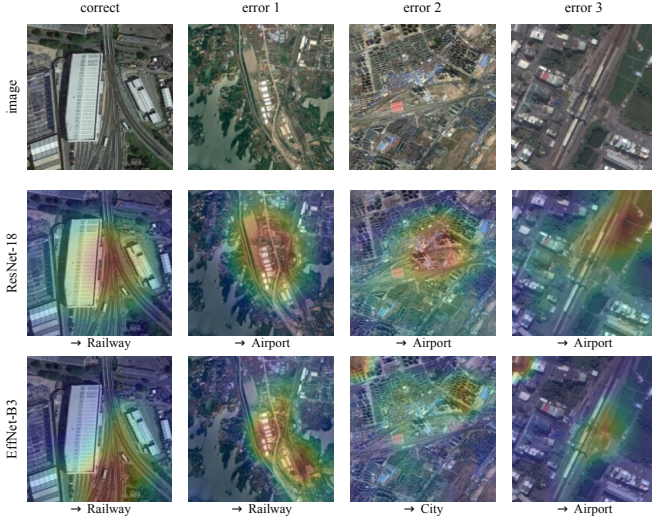


Fig. 5. When ResNet-18 mislabels *Railway* as *Airport*, its Grad-CAM focuses on large roofs or long straight strips rather than track fans. Test images (top) of the most frequent confusion after long-tailed training, with Grad-CAM maps for ResNet-18 (middle) and EfficientNet-B3 (bottom); labels give each network’s prediction.

Why augmentation hurts the CNNs. To bring every class to 544 images, the rebalanced set generates up to 15 copies of each real tail image; *River*, for example, has 34 real images and 510 synthetic ones. The hand-crafted descriptors

are histograms that barely change under rotation and mild photometric jitter, so for them the copies mostly correct the class prior. For the CNNs, one explanation consistent with our results is that they fit the repeated content of the few real tail images. Their tail recall drops below that of plain long-tailed training, whereas loss re-weighting, which shifts the prior without duplicating any content, improves both networks. The CNNs also apply random flips and rotations during training in every setting, so the offline copies add less new variation for them than for the hand-crafted pipelines. We did not try smaller augmentation budgets, and we evaluated the weighted loss for the CNNs only.

The tie-breaking pitfall. The earlier version of this study broke voting ties by member order. Every two-member ensemble therefore matched its first member exactly, which made it look as though ensembles “collapse to their weakest member”. Nothing of the sort was happening; with two voters every disagreement is a tie, and the rule decided all of them the same way. Any hard vote with an even number of members needs an explicit tie-break, and soft voting sidesteps the problem.

Limitations. All results come from one dataset with 256×256 images and 15 classes, a single (alphabetical) long-tail ordering, and one training run per configuration, so the bootstrap intervals do not capture training variance. We fine-tuned EfficientNet-B3 at 224×224 rather than its native 300×300 and unfroze only its last stage, which may understate its relative

to the fully fine-tuned ResNet-18. The Grad-CAM analysis is qualitative and covers a single confusion.

VII. CONCLUSION

We have compared hand-crafted pipelines, fine-tuned CNNs and their ensembles on SkyView, holding the test set, the validation signal and the training pool fixed and varying only the class distribution. The CNNs beat the best hand-crafted pipeline by about 12 points on balanced data and lose much less under a 16:1 long tail. For the CNNs, re-weighting the loss was the better answer to imbalance; augmenting the minority classes helped only the hand-crafted pipelines. Averaging the two CNNs gave the best result in every setting, while voting with much weaker models did not, and a two-member hard vote is only meaningful with an explicit tie-break. The obvious next steps are to repeat the study over several seeds and long-tail orderings, to try a combiner learned on validation predictions, and to check whether the findings carry over to larger benchmarks such as AID and NWPU-RESISC45 [14], [15].

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